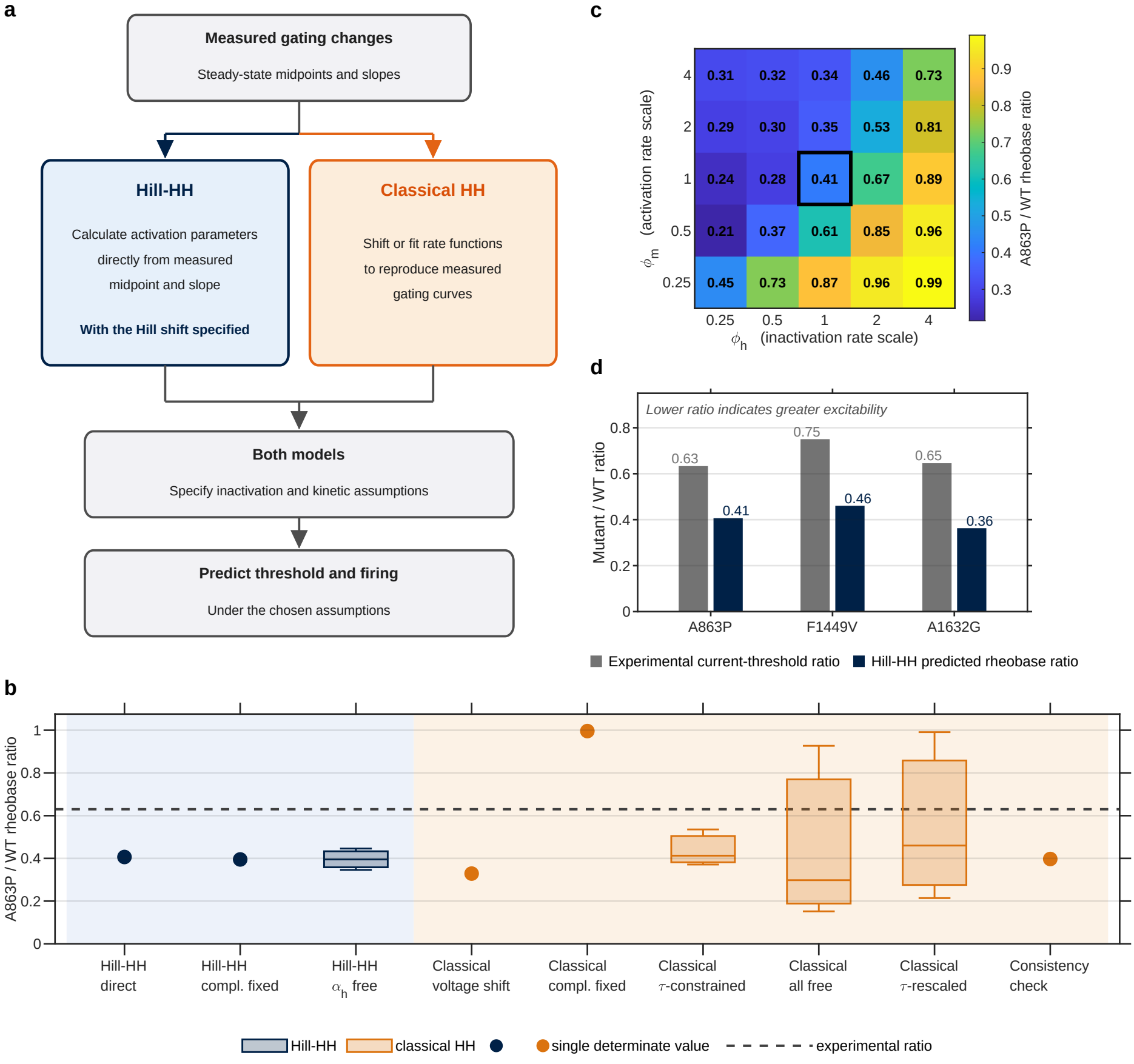
a
Measured gating changes
Steady-state midpoints and slopes
Hill-HH
Calculate activation parameters directly from measured midpoint and slope
With the Hill shift specified
Classical HH
Shift or fit rate functions to reproduce measured gating curves
Both models
Specify inactivation and kinetic assumptions
Predict threshold and firing
Under the chosen assumptions
c
0.31 0.32 0.34 0.46 0.73
0.29 0.30 0.35 0.53 0.81
0.24 0.28 0.41 0.67 0.89
0.21 0.37 0.61 0.85 0.96
0.45 0.73 0.87 0.96 0.99
4
2
1
0.5
0.25
ϕ_m (activation rate scale)
0.25 0.5 1 2 4
ϕ_h (inactivation rate scale)
A863P / WT rheobase ratio
0.9 0.8 0.7 0.6 0.5 0.4 0.3
d
Lower ratio indicates greater excitability
Mutant / WT ratio
0.8 0.6 0.4 0.2 0
0.63 0.41
0.75 0.46
0.65 0.36
A863P
F1449V
A1632G
Experimental current-threshold ratio
Hill-HH predicted rheobase ratio
b
A863P / WT rheobase ratio
1 0.8 0.6 0.4 0.2 0
Hill-HH direct
Hill-HH compl. fixed
Hill-HH α_h free
Classical voltage shift
Classical compl. fixed
Classical τ-constrained
Classical all free
Classical τ-rescaled
Consistency check
Hill-HH
classical HH
single determinate value
experimental ratio